

Smart Metro Ticket Booking System

Project Design Phase

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

1. Solution Architecture

This document walks through the end-to-end architecture of the Smart Metro Ticket Booking solution on the Now Platform — how the four personas interact with the presentation layer, how requests move through application logic and integrations, and how data and security controls underpin every layer.

1.1 Architecture Overview

Figure 7.1 — End-to-end solution architecture on the Now Platform

Passengers, Station Managers, Metro Operations, and IT Admins all reach the system through the same Service Portal / UI Pages layer, so the front end stays unified across personas. Requests are handled by the application layer — Service Catalog for input capture, Flow Designer for workflow orchestration, and Business Rules / Script Includes for fare logic — before being routed through the Integration Hub to external systems: the payment gateway, the QR code generation library, and the notification channel. All reference and transactional data lives in the data layer, and Performance Analytics turns this data into operational dashboards. Access Control Lists, OAuth 2.0, and data encryption run across every layer to keep the platform secure by design.

1.2 Component Responsibilities

Layer	Key Components	Responsibility
Presentation Layer	Service Portal, UI Pages, WhatsApp Chatbot	Single front door for all four personas across web, mobile, and chat.
Application Layer	Service Catalog, Flow Designer, Business Rules, Script Includes	Captures booking input and runs fare and workflow logic server-side.
Integration Layer	Integration Hub, REST/SOAP Spokes	Routes calls to payment, QR, and notification systems with logging and retries.
Data Layer	Station Master, Fare Rule Table, Ticket/QR Records, Transaction Logs	Single source of truth for reference and transactional data.
Reporting Layer	Performance Analytics, Reports, Dashboards	Turns stored data into usage, revenue, and peak-time insights.
Security Layer	ACLs, OAuth 2.0, Data Encryption	Enforces role-based access and protects payment and QR data end-to-end.

1.3 Design Principles

- Single platform — every persona is served from one centrally administered Now Platform instance.
- Low-code first — Flow Designer, Business Rules, and Service Catalog are favoured over custom scripting wherever feasible.
- Traceability by default — every booking, payment, and QR event carries a correlation ID.
- Security by design — ACLs and encryption apply uniformly across presentation, application, and data layers rather than being added afterward.
- Loose coupling to external systems — the Integration Hub shields the core platform from changes in the payment gateway or QR provider.